Pump makes excessive noise
Pump fails to deliver fluid
Oil leakage around pump
Pump does not deliver pressure
Low or erratic pressure
Excessive pump wear
Pump parts inside housing are broken

Pump Makes Excessive Noise

Probable Cause	Possible Remedy
A. Air leaks in the suction side of pump.	1. Install a vacuum gage in inlet on a naturally aspirated reservoir. If indicates zero, air leakage may exist. Replace fitting and/or line. 2. Replace pump shaft seal. 3. Check fluid level. 4. Check pump housing for cracks. 5. Check torque of pump housing bolts.
B. Aeration of fluid in reservoir.	1. Return lines terminate above fluid level. 2. Check fluid level. Too low causing excessive heat and foam. 3. Check fluid of foam by drawing a sample of reservoir.
C. Plugged or restricted suction line, suction strainer or filter.	1. Check for proper suction line size. 2. Check for proper filter strainer size. 3. Remove, clean or replace filter or strainer.
D. Plugged reservoir filer/ breather filler neck strain.	1. Remove, clean or replace.
E. Fluid viscosity too high.	1. Drain system and fill with fluid recommended by manufacturer.
F. Worn or stuck pump parts.	1.Check for burrs or solid contaminants. If burrs, remove, clean and/or replace parts as necessary. If contaminated, flush system thoroughly and filter fluid at refill. 2. Pump housing bolts improperly torqued. Loosen and re-torque to proper specifications.
G. Improper installation.	1. Check alignment with drive mechanism. 2. Check vane cartridge for "backward" vanes. 3. Check rotation. 4. Check pump rpm. 5. Check relief valve or unloading valve for proper setting.

CFC INDUSTRIAL TRAINING

Pump Fails to Deliver Fluid

Probable Cause	Possible Remedy
A. Lower fluid level in reservoir.	1. Check fluid level, fill to proper level.
B. Pump intake restricted or plugged.	1. Check for proper suction line size. 2. Check for proper filter or strainer size. 3. Remove, clean or replace filter and/or strainer.
C. Air leak in suction line and/or air lock.	1. Repair leaks. 2. Check pump housing drain pot for improper positioning, allowing air to be trapped by housing. 3. Bleed pump inlet.
D. Pump shaft turning slowly.	1. Check prime mover's rpm. Increase speed to recommended specifications. 2. If belt driven (mobile), check belt tension.
E. Oil viscosity too high or too low.	1. Drain, flush and fill with filtered fluid recommended by pump manufacturer.
F. Pump shaft or parts broken or worn.	1. Remove and replace broken or worn parts.
G. Dirt or sludge in the pump.	1. Disassemble and clean pump. 2. Flush system thoroughly and fill with clean, filtered fluid.
H. Improperly adjusted variable volume adjustment mechanism.	1. Check and set adjustment properly.
I. Improperly adjusted pressure compensator.	1. Check pressure setting and set properly.
J. Wrong rotation.	1. Check rotation on pup and prime mover.

CFC INDUSTRIAL TRAINING

Oil Leakage Around Pump

Probable Cause	Possible Remedy
A. Shaft seal worn.	1. Remove and replace. 2. Check shaft for scratches and grooves. Replace if necessary.
B. Suction or pressure line connection's loose or broken.	1. Tighten or replace connections.
C. Pump housing bolts are loose or have improper torque.	1. Disassemble and inspect seals for damage. 2. Re-torque to specification. Follow manufacturer's torquing procedure.
D. Case drain line too small or restricted (shaft seal leaking)	1. Check for proper line and fitting size. 2. Eliminate kinks and bends. 3. Check ease pressure against manufacturer specifications.
E. Cracked housing over pressurized.	1. Check relief valve pressure settings. 2. Replace pump. 3. Follow pump manufacturer's recommendations for pump start-up procedure.

CFC INDUSTRIAL TRAINING

Pump Does Not Deliver Pressure

Probable Cause	Possible Remedy
A. Pump not delivering fluid.	1. See corresponding troubleshooting chart.
B. Vane(s) in vane pump sticking. Piston(s) in piston pump sticking.	1. Disassemble and inspect for burrs and/or varnishing. Replace and/or repair as necessary. 2. Flush system thoroughly and refill with filtered fluid.
C. System relief valve unloading. (compensator malfunction)	1. Check each broken, worn or stuck parts. If stuck due to contamination, drain, flush and refill system.

CFC INDUSTRIAL TRAINING

Low or Erratic Pressure

Probable Cause	Possible Remedy
A. Cold fluid.	1. Operate machine to raise temperature of fluid to normal operating temperature. Check pressure again.
B. Wrong fluid viscosity.	1. Drain and refill system with filtered recommended fluid.
C. Aeration or cavitation.	1. See excessive noise troubleshooting.
D. Excessive pump wear.	1. See corresponding troubleshooting chart.
E. Sticking pump parts.	1. Disassemble and repair as necessary. 2. Check for varnish and/or sludge. 3. Clean parts thoroughly, flush and filter entire system.
F. Pump speed too slow.	1. Check rpm and adjust speed, but do not exceed pump specifications.

CFC INDUSTRIAL TRAINING

Excessive Pump Wear

Probable Cause	Possible Remedy
A. Abrasive dirt in the hydraulic fluid.	1. Drain and flush system thoroughly. 2. Replace filter elements.
B. High water content fluid rating.	1. Check manufacturer's recommendation on use of High Water Content Fluid (HWCF) in the pump.
C. System pressure exceeding pumps rating.	1. Check for possible relief malfunction. 2. Check for pressure regulator failures.
D. Oil viscosity too high or too low.	1. Drain, flush and refill system with proper viscosity fluid.
E. Pump misalignment or belt drive too tight.	1. Check pump alignment with motor and adjustment needed. 2. Check belt tension. 3. Check to thrust loads.
F. Aeration and cavitation.	1. See excessive noise troubleshooting chart. 2. Check parts for the severity of wear and replace if necessary.

CFC INDUSTRIAL TRAINING

Pump Parts Inside Housing are Broken

Probable Cause	Possible Remedy
A. Seizure due to lack of fluid.	1. Check for cavitation of aeration. 2. Check for low or too high an oil level. 3. Check viscosity of fluid.
B. Excessive system pressure above maximum pump rating.	1. Check pressure controls for possible malfunctions. Repair or place as necessary.
C. Excessive torquing of housing bolts.	1. Repair or replace parts as necessary. 2. Reassemble following manufacturer's procedures.
D. Solid matter contamination being drawn in from the reservoir seizing pump parts.	1. Check suction strainer for damage. Repair and replace. 2. Check suction filter bypass condition. 3. Disassemble pump, repair and/or replace parts as needed. 4. Clean up fluid in system by filtering through a filter cart system or by adding filters to the system.

CFC INDUSTRIAL TRAINING